

Transformations Lab

You need: tracing paper (or patty paper), a straightedge, a protractor, and graph paper. Everything you are about to discover, you will later prove with coordinates — this is where the rules come from.

Lab 1 - Is a Translation Really a Rigid Motion?

Show that a translation preserves both segment length and angle measure.

Step 1. On graph paper, draw $\triangle ABC$ with $A(-5, 1)$, $B(-2, 4)$, and $C(-1, 0)$. Trace it onto tracing paper.

Step 2. Slide the tracing paper 6 units right and 3 units down — do NOT turn or flip it. Press through the vertices to mark A' , B' , and C' .

Step 3. Measure AB and $A'B'$ with a ruler. Measure $\angle A$ and $\angle A'$ with a protractor.

- ▶ What did you find about AB and $A'B'$? About $m\angle A$ and $m\angle A'$?
- ▶ Write the translation as a rule: $(x, y) \rightarrow (\rule{1.5cm}{0.4pt}, \rule{1.5cm}{0.4pt})$.
- ▶ Measure AA' , BB' , and CC' . What do you notice? Why does that happen?

- A transformation that preserves length AND angle measure is called a rigid motion. Because a translation is a rigid motion, the image is always congruent to the preimage.

Lab 2 - Where the Reflection Rules Come From

Fold, do not calculate. Then write the coordinate rule you discovered.

Step 1. Plot $P(4, 3)$ on graph paper. Fold the paper along the x -axis and press P through the fold. Label the image and record its coordinates.

Step 2. Unfold. Now fold along the y -axis and press P through. Record those coordinates.

Step 3. Unfold. Draw the line $y = x$, fold along it, and press P through. Record those coordinates.

Step 4. Draw PP' for each fold. Measure the angle where PP' crosses the fold line, and compare the two halves of PP' .

- ▶ The fold line is the _____ of every segment joining a point to its image.
- ▶ Complete the rules: x -axis: $(a, b) \rightarrow (\rule{1cm}{0.4pt}, \rule{1cm}{0.4pt})$ y -axis: $(a, b) \rightarrow (\rule{1cm}{0.4pt}, \rule{1cm}{0.4pt})$ $y = x$: $(a, b) \rightarrow (\rule{1cm}{0.4pt}, \rule{1cm}{0.4pt})$

Lab 3 - Turning: Where the Rotation Rules Come From

Use tracing paper to turn a figure and read off the coordinate rule.

Step 1. Plot $Q(5, 2)$ on graph paper and mark the origin O . Trace both onto tracing paper.

Step 2. Hold your pencil point firmly on the origin and turn the tracing paper counterclockwise until the axes line up again — that is a 90° turn. Press Q through and record Q' .

Step 3. Turn another 90° (180° total from the start) and record Q'' . Turn one more time (270°) and record Q''' .

Step 4. Draw OQ and OQ' . Measure $\angle QOQ'$ with a protractor, then measure OQ and OQ' .

- ▶ What is $m\angle QOQ'$? What do you notice about OQ and OQ' ? Why does that make a rotation a rigid motion?
- ▶ Complete the rules: $90^\circ: (a, b) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$ $180^\circ: (a, b) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$ $270^\circ: (a, b) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
- ▶ Try a clockwise 90° turn. Which counterclockwise rotation gives the same image? Explain.

Lab 4 - Build the Rule Table

Fill in the table from the rules you discovered. The last row is the whole point — it is the rule itself.

Point	In the x-axis	In the y-axis	In $y = x$	90° about O	180° about O
$(3, 5)$					
$(-2, 4)$					
$(6, -1)$					
(a, b)					

- ▶ Two of these six columns give the SAME image for every point. Which two, and why?
- ▶ Which transformations in the table are rigid motions? How can you tell without measuring?

Lab 5 - Symmetry Hunt

Fold and turn to find symmetry, then say how many of each a figure has.

Step 1. Cut out a regular hexagon, a rectangle that is not a square, and an isosceles trapezoid.

Step 2. Fold each figure every way you can so the two halves match exactly. Count the folds — those are the lines of symmetry.

Step 3. Now put a pin through the center of each figure and turn it. Count every turn of 180° or less that lands the figure back on itself.

- ▶ Fill in: hexagon $\rightarrow$ _____ lines, rotations of _____; rectangle $\rightarrow$ _____ lines, rotations of _____; trapezoid $\rightarrow$ _____ lines, rotations of _____.
- ▶ Name a figure with rotational symmetry but NO line symmetry.